Master Rehearsal Guide

Skin Lesion Classification Thesis

Everything you need to defend successfully

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USTH ICT 2026

Contents

1. How to use this document

Orientation

This is the **single file** you should read while rehearsing. It covers:

- Every technique in the project — explained *plainly* first, then with *rigorous math*.
- Every updated number through Week 3 (multi-seed, metadata fusion, OOD).
- Every concept paired with a *concrete worked example*.
- The 10 hardest questions the committee may ask + answer templates.
- Defense-day checklist.
- A one-page summary you can print and carry.

Colour-coded boxes:

Plain-English Explanation

Green box — everyday-language explanation, no formulas. Understand *this* before going into the math.

Example

Yellow box — concrete numerical example using actual project numbers. This is how you answer "give me an example" during Q&A.

Blue box — key insight / deep understanding

The single most important bullet of a section. Re-read these before defense.

Committee may ask

Orange box — hypothetical committee question with a model answer.

Suggested reading schedule:

- Round 1 (90 min): read sequentially from section 2 to section 22.
- Round 2 (45 min): only the blue and orange boxes.
- Round 3 (15 min on the morning of defense): only section 23 (one-page summary).

2. What is this thesis — 1-page elevator pitch

Problem. Given one dermoscopic image (a skin lesion photographed under a polarised-light dermoscope), predict **which of 7 lesion classes** it belongs to: `akiec`, `bcc`, `bkl`, `df`, `mel`, `nv`, `vasc`. Among these, `mel` (melanoma) is the **most dangerous**.

Data. ISIC 2018 Task 3 / HAM10000 — 10,015 images, severe imbalance (NV is 67%, DF is only 1.1%, ratio **58:1**).

Method. Fine-tune 4 ImageNet-pretrained models (*EfficientNet-B0, ResNet50, DenseNet121, Swin-Tiny*) on ISIC 2018, combine them with **soft-voting ensemble**, and extend with **patient metadata fusion** (age, sex, anatomical localisation).

Headline results:

- Best single model (seed 42): EfficientNet-B0 — Acc 88.36%, F1 0.831.
- Image-only ensemble (1 seed): Acc **89.49%**, F1 **0.845**, AUC **0.980**.
- Image-only ensemble (5 seeds): Acc **89.93** $\pm$ **0.66%**, F1 **0.852** $\pm$ **0.010**, AUC **0.982** $\pm$ **0.001**.
- Ensemble + metadata fusion: Acc **90.49%**, F1 **0.873**, AUC **0.984** — beats Pacheco & Krohling 2020.
- Out-of-distribution on PAD-UFES-20 (smartphone photos, zero-shot): Acc **32.46%** — a 56 pp drop, the standard cost of dermoscopy-to-smartphone transfer.

Interpretability. Grad-CAM (Selvaraju 2017) + 4 quantitative metrics: `focus_ratio > 0.6` and `peak_distance < 0.2` across all 4 models ⇒ the model actually attends to the lesion, not to artefacts.

Product. Streamlit prototype (`app.py`) that lets clinicians upload an image and receive a diagnosis + heatmap in real time.

One-line take-away

A carefully regularised pipeline, an architecturally diverse ensemble, plus patient metadata can reach 90%+ on ISIC 2018 and beat several published SOTA — but real-world deployment on smartphone photographs still requires domain adaptation (the 56 pp OOD drop bounds this honestly).

3. Foundational definitions

3.1. Deep learning

Plain-English Explanation

Deep learning = a neural network with *many layers* stacked on top of each other. Each layer takes the previous layer's output and transforms it into a *more abstract* representation. For images: the first layers see *edges*, middle layers see *parts*, final layers see *concepts*.

Mathematically: given data (x_i, y_i) and parameters θ, the network learns by gradient descent:

$$\theta^* = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\big(f_\theta(x_i),\, y_i\big).$$

3.2. Pre-training — training "from scratch"

Plain-English Explanation

Pre-training = teach a network from zero on a *very large* dataset. The goal: teach the network how to "see" in general. Example: teach the network to distinguish cats, dogs and cars — it knows nothing about dermatology yet.

Pre-training on ImageNet

ImageNet-1K has 1.28 *million* images across 1,000 classes. Pre-training EfficientNet-B0 on ImageNet takes *weeks* on many GPUs. The output: a weights file θ_{pre} — the network "knows how to look at images" but knows nothing about skin yet.

3.3. Fine-tuning — "small adjustment"

Plain-English Explanation

Fine-tuning = take a pre-trained network, change its output for a new task, then keep training on a smaller dataset. Like: you already know English, now you study *medical* English — no need to start from A-Z, just learn the specialised vocabulary.

Fine-tuning EB0 for ISIC 2018

1. Load θ_{pre} (ImageNet weights) from `timm`.
2. Replace the classifier head: `Linear(1280, 1000)` $\rightarrow$ `Linear(1280, 7)` (random init).
3. Train the entire network on 7,010 ISIC images with a small LR of 3e-4 for 50 epochs.

When to use which?

Training from scratch needs $>$ 100K images and weeks of GPU time — not feasible on the 10K-image ISIC dataset. **Fine-tuning** needs a few thousand images and minutes-to-hours of training — this is exactly what the thesis does.

3.4. Which one does this thesis use?

The thesis uses fine-tuning. It loads ImageNet weights from `timm` and does *not* pre-train anything itself. The actual code:

Listing 1: `src/models/build_model.py`

```python
import timm
def build_model(model_cfg: dict):
    model = timm.create_model(
        model_cfg['name'],       # "tf_efficientnet_b0", "resnet50", ...
        pretrained=True,         # load ImageNet weights
        num_classes=7,           # replace head: 1000 -> 7
        drop_rate=0.3,           # dropout before classifier
    )
    return model
```

Committee may ask

Q: Did you pre-train or fine-tune?
A: Fine-tuning. I do not have the data to pre-train from scratch (that needs millions of images). I load ImageNet-pretrained weights from `timm`, replace the classifier head from 1000 ImageNet classes down to 7 ISIC classes, then train the entire network on ISIC with a small learning rate.

4. The ISIC 2018 / HAM10000 dataset

4.1. The seven lesion classes

Code	Full name	Train	Val	Test	Total
nv	Melanocytic Nevus (common mole)	4,693	1,006	1,006	6,705
mel	**Melanoma**	779	167	167	1,113
bkl	Benign Keratosis-like	769	165	165	1,099
bcc	Basal Cell Carcinoma	360	77	77	514
akiec	Actinic Keratosis (pre-cancerous)	229	49	49	327
vasc	Vascular Lesion	99	21	22	142
df	Dermatofibroma	81	17	17	115
Total		7,010	1,502	1,503	**10,015**

4.2. Extreme class imbalance

Plain-English Explanation

The largest class (NV: 4,693 images) versus the smallest (DF: 81 images) gives a ratio of **58:1**. Equivalently, in 59 random images you would expect 58 NV and 1 DF. This imbalance naturally biases any model toward NV.

Why is imbalance dangerous?

A *lazy* model could predict "NV for everything" and reach **67% accuracy** without learning anything — but it would miss 100% of melanomas. That is why we report **Macro F1** (the *unweighted* mean of per-class F1) as the headline metric, not accuracy.

4.3. Stratified 70/15/15 split

Plain-English Explanation

Stratified = split the data so that the 7-class proportions are preserved across train/val/test. A purely random split risks leaving the val set with zero DF samples (only 17 exist in 10K).

5. Pre-processing + augmentation

5.1. Base pre-processing

Every image is:

- **Resized to 224×224** — the standard ImageNet input size.
- **ToTensor** — pixels $[0, 255] \to [0, 1]$.
- **Normalised** with ImageNet mean/std:

$$x' = \frac{x - \mu_{\text{ImageNet}}}{\sigma_{\text{ImageNet}}}, \quad \mu = (0.485, 0.456, 0.406), \quad \sigma = (0.229, 0.224, 0.225).$$

Plain-English Explanation

Why use ImageNet's mean and std? The pretrained model has already learned to expect inputs at that statistical scale. Normalising differently shifts the input distribution away from training distribution $\Rightarrow$ the model panics.

5.2. Augmentation (train only, never on val/test)

Transform	Parameter	Reason
RandomResizedCrop	scale 0.85–1.0	Scale invariance
HorizontalFlip	$p = 0.5$	Skin images are symmetric
VerticalFlip	$p = 0.5$	No fixed “up/down” orientation
Rotation	$\pm 90°$	Rotation invariance
ColorJitter	b/c/s/h $= 0.25$	Robustness to device
GaussianBlur	$p = 0.1$	Robustness to sharpness
RandomErasing	$p = 0.1$	Robustness to occlusion

6. Weighted Random Sampler

6.1. Idea

Plain-English Explanation

A vanilla DataLoader picks images uniformly $\rightarrow$ a single batch is mostly NV. The Weighted Sampler assigns a *higher weight* to rare classes $\rightarrow$ the DataLoader picks DF more often $\rightarrow$ each batch is approximately class-balanced.

6.2. Formula

$$w_c = \frac{1}{n_c} \quad \text{(or normalised)} \quad w_c = \frac{N}{C \cdot n_c}$$

Per-class weights on ISIC train

$N = 7{,}010$, $C = 7$:

Class	n_c	w_c
nv	4,693	0.213
mel	779	1.286
bkl	769	1.303
bcc	360	2.782
akiec	229	4.373
vasc	99	10.117
df	81	**12.365**

Consequence: a DF image is sampled with probability $12.365/0.213 \approx$ **58**$\times$ that of an NV image. After sampling, each mini-batch is roughly class-balanced.

7. Mixup — "blend two images"

Plain-English Explanation

Mixup = take two different images, *blend* them at a random ratio, and blend their labels at the same ratio. This forces the model to behave *linearly* between samples and prevents memorisation.

7.1. Formula

Given two samples (x_i, y_i) and (x_j, y_j), draw $\lambda \sim \text{Beta}(\alpha, \alpha)$ with $\alpha = 0.2$:

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j, \qquad \tilde{y} = \lambda y_i + (1-\lambda)y_j.$$

Mixup of a melanoma and a nevus

- x_i = a melanoma image, $y_i = [0, 0, 0, 0, \mathbf{1}, 0, 0]$ (mel)
- x_j = a nevus image, $y_j = [0, 0, 0, 0, 0, \mathbf{1}, 0]$ (nv)
- $\lambda = 0.7$ drawn at random.

Blended image: $\tilde{x} = 0.7\, x_i + 0.3\, x_j$ — the melanoma image is "70% clear" and a faint "30% nevus" is overlaid on it.
Blended label: $\tilde{y} = [0, 0, 0, 0, \mathbf{0.7}, \mathbf{0.3}, 0]$.
The model must predict probability mel near 0.7 *and* nv near 0.3.

Ablation: Mixup is the most important component

Removing Mixup/CutMix from the pipeline drops Macro F1 by **4.51%** — the largest single contribution of any component. With only 81 DF training images, without Mixup the model effectively *memorises* those 81 images in the first few epochs.

8. CutMix — "cut and paste a patch"

Plain-English Explanation

CutMix differs from Mixup: instead of blending the whole image (which makes a blurry overlay), CutMix *cuts a rectangular patch* from image j and *pastes* it onto image i. The blended label follows the *area* of the patch.

8.1. Formula

$$\tilde{x} = M \odot x_i + (1-M) \odot x_j, \qquad \tilde{y} = \lambda y_i + (1-\lambda)y_j,$$

where $M \in \{0,1\}^{H \times W}$ is a bounding-box mask and $\lambda = 1 - \frac{\text{patch area}}{H \cdot W}$.

CutMix with an 80×80 patch

- x_i = a nevus image 224×224, y_i = nv.
- x_j = a melanoma image 224×224, y_j = mel.
- Sample a random bounding box of size 80×80 in the top-left corner.

Blended image: the nevus image, but the 80×80 region is replaced by pixels from the melanoma.
λ: $1 - 80 \cdot 80/(224 \cdot 224) = 1 - 0.128 = 0.872$.
Blended label: $[0, 0, 0, 0, \mathbf{0.128}, \mathbf{0.872}, 0]$.

9. Label Smoothing — "soften the labels"

Plain-English Explanation

A one-hot label says "*100% certain* this is MEL, 0% everything else". Label smoothing replaces it with "**91%** MEL, 1.5% each of the other classes". This prevents the model from becoming overconfident.

9.1. Formula

$$y_k^{\text{LS}} = (1 - \varepsilon)\mathbb{1}[k = c] + \frac{\varepsilon}{C}, \quad \varepsilon = 0.1,\ C = 7.$$

Smoothing the MEL label

$y_{\text{onehot}} = [0, 0, 0, 0, 1, 0, 0]$ (mel).
With $\varepsilon = 0.1$, $C = 7$: the "correct" part shrinks to $1 - 0.1 = 0.9$, and 0.1 is spread evenly across the 7 classes: $0.1/7 \approx 0.0143$.

$$y^{\text{LS}} = [\mathbf{0.0143}, 0.0143, 0.0143, 0.0143, \mathbf{0.9143}, 0.0143, 0.0143].$$

Sanity check: $0.9143 + 6 \cdot 0.0143 = 1.000$. □

Observed paradox

Removing Label Smoothing *raises* AUC (0.952 → 0.967) but *lowers* Macro F1 by 3.11%. Hard targets ⇒ sharper probability ranking (higher AUC), but harder decision boundaries make more errors at the boundary regions (lower F1).

10. The four model architectures

10.1. EfficientNet-B0 (5.3M parameters)

Plain-English Explanation

The *smallest* of the four models. Its idea: simultaneously scale up the network's *depth* (number of layers), *width* (number of channels) and *resolution* (input image size) using a single shared scaling coefficient — this is called "compound scaling". It also has Squeeze-and-Excitation (SE) blocks that learn channel-wise global context.

10.2. ResNet50 (25.6M parameters)

Plain-English Explanation

A *classical* architecture. Core idea: the residual connection $f(x) = F(x) + x$. This lets very deep networks (50+ layers) train without vanishing gradients.

10.3. DenseNet121 (8.0M parameters)

Plain-English Explanation

Each layer receives the concatenation of *every* previous layer — this is called dense connectivity. It reuses already-learned features and keeps the parameter count down.

10.4. Swin-Tiny (28.0M parameters, Vision Transformer)

Plain-English Explanation

Substantially *different* from the three CNNs above. It uses *self-attention* instead of convolution. The image is split into small windows, attention is computed within each window, then the windows are *shifted* so different regions can exchange information. The architecture is hierarchical with 4 stages.

Why pick four diverse architectures?

Three CNNs (EB0, RN50, DN121) + one Transformer (Swin) $\Rightarrow$ *uncorrelated errors*. CNNs focus on local features, Swin focuses on global context. When you ensemble them, one model's mistakes do not overlap with another's $\Rightarrow$ averaging the probabilities actually improves accuracy.

11. Full training pipeline

11.1. Loss function

Cross-entropy with label smoothing:

$$\mathcal{L}_{\text{CE}}(p, y) = -\sum_{k=1}^{C} y_k^{\text{LS}} \log p_k.$$

11.2. Optimiser: AdamW

Plain-English Explanation

Adam = a "smart" optimiser that automatically adjusts the learning rate for each parameter. AdamW = Adam with weight decay applied *correctly* (directly on the parameters, not added to the gradient). It is the modern default.

$$\theta_t = \theta_{t-1} - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_{t-1} \right).$$

11.3. Cosine annealing with warmup

Plain-English Explanation

Warmup: LR rises from near 0 to a peak over the first 3–5 epochs — avoids destroying the pretrained weights right at the start. Cosine: LR then decays along a cosine curve from peak down to a minimum — a gentle convergence at the end.

11.4. Hyperparameters

	EB0	**RN50**	**DN121**	**Swin**
Learning rate	3e-4	3e-4	2.5e-4	**1e-4**
Weight decay	1e-3	1e-3	1e-3	**5e-2**
Warmup epochs	3	3	3	5
Max epochs	50	50	50	50
Batch size	16	16	16	16
Mixup / CutMix α	0.2/1.0	0.2/1.0	0.2/1.0	0.2/1.0
Label smoothing ε	0.1	0.1	0.1	0.1

Why is Swin configured differently?

Transformers lack the inductive bias of CNNs $\Rightarrow$ they need a **lower LR** to avoid destroying the pre-trained attention weights, and a **higher weight decay** to limit overfitting (28M parameters on 7K images is a very overfit-prone ratio).

12. Soft-voting ensemble

12.1. Why ensemble?

Plain-English Explanation

A single model tends to be wrong on a *specific subset* of images. Another model is wrong on *a different subset*. Averaging four models cancels individual errors and keeps the consensus signal.

12.2. Soft-voting formula

With $M = 4$ models and each model m producing a probability vector $p_m(x) \in [0, 1]^7$:

$$\bar{p}(x) = \frac{1}{M} \sum_{m=1}^{M} p_m(x), \qquad \hat{y} = \arg\max_c \bar{p}_c(x).$$

Soft voting on one real test image

The four models produce these probability vectors over the 7 classes (akiec, bcc, bkl, df, **mel**, nv, vasc):

	akiec	bcc	bkl	df	**mel**	nv	vasc
EB0	0.02	0.03	0.05	0.01	**0.65**	0.22	0.02
RN50	0.03	0.05	0.08	0.01	**0.55**	0.26	0.02
DN121	0.02	0.03	0.06	0.01	**0.62**	0.24	0.02
Swin	0.04	0.04	0.05	0.01	**0.48**	0.36	0.02
Mean	0.028	0.038	0.060	0.010	**0.575**	0.270	0.020

Argmax: `mel` with ensemble probability 57.5%. Swin is "not very confident" (48%) but the CNNs lift the average up — the ensemble remains robust.

12.3. Why soft voting over hard voting or stacking?

- **Hard voting** (each model takes argmax then votes): discards probability information. Cannot distinguish "all four are sure of MEL" from "two say MEL 51%, two say NV 49%".
- **Stacking** (a meta-learner learns ensemble weights): requires a second validation set, easy to overfit on small data.
- **Soft voting**: simple, stable, no extra training — already reaches 89.49% in-distribution and there is no clear headroom for stacking to overtake.

13. Multi-seed + McNemar (Week 1)

13.1. Why multi-seed?

Plain-English Explanation

One random seed = one "lucky roll". The committee will ask: "how do you know 89.49% is not luck?" The answer: rerun the experiment with 5 different seeds and report *mean* $\pm$ *standard deviation*.

13.2. Five-seed results

Model	**Accuracy**	**Macro F1**	**Macro AUC**
EfficientNet-B0	$87.31 \pm 1.32\%$	0.814 ± 0.022	0.965 ± 0.006
ResNet50	$86.31 \pm 0.57\%$	0.792 ± 0.011	0.949 ± 0.003
DenseNet121	$85.62 \pm 2.42\%$	0.797 ± 0.025	0.960 ± 0.004
Swin-Tiny	$\mathbf{88.33 \pm 1.96\%}$	$\mathbf{0.839 \pm 0.030}$	0.966 ± 0.008
Ensemble	$\mathbf{89.93 \pm 0.66\%}$	$\mathbf{0.852 \pm 0.010}$	$\mathbf{0.982 \pm 0.001}$

Three important findings

(1) The 89.49% (single seed) lies within 1σ of the 89.93% mean $\Rightarrow$ **not lucky, statistically robust**.
(2) Swin-Tiny is in fact the strongest single backbone when averaged across seeds (88.33% > EB0's 87.31%). Seed 42 was simply a bad draw for Swin.
(3) The ensemble's standard deviation is tiny (Acc 0.66%, AUC 0.001) $\Rightarrow$ soft voting absorbs seed-level variance.

13.3. McNemar's test — statistical proof

Plain-English Explanation

McNemar = a statistical test comparing two classifiers on the *same test set*. It counts: on how many images is A right and B wrong vs. on how many is A wrong and B right. If the gap is large ⇒ one classifier is *genuinely* better than the other, not by chance.

The McNemar formula (with continuity correction):

$$\chi^2 = \frac{(|n_{ab} - n_{ba}| - 1)^2}{n_{ab} + n_{ba}}, \quad p = \chi^2_1\text{-survival}(\chi^2).$$

McNemar: Swin vs. Ensemble, seed 42

1503 test images:

- $n_{ab} = 33$ (Swin correct, ensemble wrong)
- $n_{ba} = 93$ (Swin wrong, ensemble correct)
- $\chi^2 = (|33 - 93| - 1)^2/(33 + 93) = 59^2/126 = 27.63$
- $p = 1.47 \times 10^{-7}$

$p \ll 0.05$ ⇒ the ensemble is **genuinely** better than Swin, not by chance.

13.4. Combining across 5 seeds with Fisher's method

$$\chi^2_{\text{combined}} = -2\sum_{i=1}^{k} \log p_i, \quad df = 2k.$$

Single model vs. Ensemble	Ensemble wins	Fisher combined p
EfficientNet-B0	5/5	5.3×10^{-18}
ResNet50	5/5	4.5×10^{-31}
DenseNet121	5/5	6.7×10^{-39}
Swin-Tiny (strongest)	**5/5**	$\mathbf{1.2 \times 10^{-7}}$

14. Grad-CAM — interpretability

14.1. Why is Grad-CAM useful?

Plain-English Explanation

The model says "MEL 90%" but *why*? Grad-CAM produces a heatmap: the redder a region, the more the model relied on it to make the decision. If the model attends to the lesion ⇒ trustworthy. If it attends to hair / ruler / image corners ⇒ less so.

14.2. Four mathematical steps

Given image x, target class c, and feature map $A \in \mathbb{R}^{K \times H \times W}$ at the last convolutional layer:

Step 1 — forward + backward:

$$y^c = f_\theta(x)[c], \qquad \frac{\partial y^c}{\partial A^k_{ij}}.$$

Step 2 — channel weights (GAP over gradients):

$$\alpha^k_c = \frac{1}{HW} \sum_{i,j} \frac{\partial y^c}{\partial A^k_{ij}}.$$

Step 3 — heatmap (ReLU):

$$L_c = \mathrm{ReLU}\left(\sum_k \alpha^k_c \, A^k \right).$$

Step 4: upsample to 224×224, normalise to $[0, 1]$, overlay with the JET colormap.

14.3. Target layer per model

Model	Target layer	Shape
EfficientNet-B0	`model.blocks[-1][-1]`	(320, 7, 7)
ResNet50	`model.layer4[-1]`	(2048, 7, 7)
DenseNet121	`model.features.norm5`	(1024, 7, 7)
Swin-Tiny	`model.layers[-1].blocks[-1].norm2`	(49, 768)

The `model.bn2` debug story

The first attempt used `model.bn2` on EfficientNet (BatchNormAct2d + SiLU). Because $\mathrm{SiLU}(x) \approx 0$ for $x < 0$, about 50% of activations zero out after BN $\Rightarrow$ only 18% of pixels carry meaningful activation $\Rightarrow$ heatmaps are *uniformly blue*, with no focus.
Switching to `blocks[-1][-1]` gives 51% active pixels and heatmaps that *correctly outline the lesion*.
Without quantitative metrics, this error would have slipped past visual inspection.

14.4. Quantitative Grad-CAM (4 metrics)

`focus_ratio`	Fraction of total activation inside the lesion mask (Otsu). High = correct focus.
`peak_distance`	Distance from peak activation to lesion centroid. Low = focus on the centre.
`entropy`	Shannon entropy of the heatmap. Low = sharp focus.
`coverage_75`	% area containing the top-25% activation. Low = compact.

All 4 models achieve `focus_ratio` > 0.6 and `peak_distance` < 0.2 on the test set.

15. Evaluation metrics

15.1. Why accuracy alone is not enough

Plain-English Explanation

On 58:1 imbalanced data, "predict NV for every image" gives 67% accuracy while missing 100% of melanomas. So accuracy is a *misleading* metric.

15.2. Definition of the headline metrics

Metric	Formula	Meaning
Accuracy	$\sum TP_c/N$	Overall correctness (biased on imbalanced data)
Macro F1	$\frac{1}{C}\sum F1_c$	**Unweighted mean of per-class F1 — HEADLINE METRIC**
Weighted F1	$\sum F1_c \cdot n_c/N$	F1 weighted by class size
Macro AUC	$\frac{1}{C}\sum \mathrm{AUC}_c$	Probability-ranking capability
Cohen's κ	$(P_o - P_e)/(1 - P_e)$	Agreement above chance
MCC	most balanced	Uses all four confusion cells
Sensitivity	$TP/(TP + FN)$	Recall — catching positives correctly
Specificity	$TN/(TN + FP)$	Identifying negatives correctly

Ensemble's melanoma F1

167 true MEL images in the test set. The ensemble predicts "MEL" on 166 images, of which 124 are actually MEL.

$$TP = 124, \quad FP = 166 - 124 = 42, \quad FN = 167 - 124 = 43$$
$$\text{Precision} = 124/166 = 0.747$$
$$\text{Recall} = 124/167 = 0.740$$
$$F1 = \frac{2 \cdot 0.747 \cdot 0.740}{0.747 + 0.740} = \mathbf{0.744}$$

16. Ablation study

16.1. Methodology

Plain-English Explanation

Remove one component at a time from the pipeline, keep everything else fixed, measure the drop in Macro F1. The component whose removal causes the largest drop is the most important one.

16.2. Results

16.3. Three findings and two paradoxes

1. Mixup/CutMix is the most important (-4.51% F1). With 81 training DF images, without Mixup the model memorises the rare samples.

2. Label-smoothing paradox. Removing LS *raises* AUC ($0.952 \to 0.967$) but *lowers* F1 by 3.11%. Hard targets $\Rightarrow$ sharper rankings (higher AUC), but harder boundaries hurt MEL/NV/BKL (lower F1).

3. Augmentation paradox. Removing strong augmentation actually *raises* accuracy (88.96% > 88.36%)

Configuration	Acc	Macro F1	AUC	Δ F1
Full pipeline	**0.8836**	**0.8311**	0.9522	—
– Mixup & CutMix	0.8589	0.7860	0.9486	$-$**0.0451**
– Label Smoothing	0.8776	0.8000	**0.9674**	-0.0311
– Weighted Sampler	0.8762	0.8041	0.9521	-0.0270
– Advanced Aug	**0.8896**	0.8204	0.9664	-0.0107

because the model fits NV more easily. But Macro F1 drops — minority classes are abandoned.

Core lesson

The largest contribution does **not** come from a bigger architecture, it comes from **better regularisation**. Mixup/CutMix + LS + Weighted Sampler together contribute roughly $\sim$**10% F1** — more than the spread between architectures.

17. Metadata fusion (Week 2) — crossing 90%

17.1. Idea

Plain-English Explanation

Images alone are not enough. A clinician always considers age, sex and anatomical location before diagnosing. A brown lesion on the back of a 30-year-old is very different from the same lesion on the face of a 75-year-old. The image-only pipeline ignores all of this. In Week 2 I added a 19-dimensional metadata vector to the pipeline.

17.2. The 19-dimensional metadata vector

- 1 dim: standardised age (mean 51.86, std 16.97, missing $\to$ mean).
- 3 dims: one-hot sex (male / female / unknown).
- 15 dims: one-hot anatomical localisation (abdomen, back, chest, ear, face, foot, ..., unknown).

17.3. Late-fusion architecture

$$\text{logits} = \text{Linear}\big(\text{Concat}(f_{\text{img}}(x), \text{MLP}(\text{meta}))\big)$$

The MLP has two layers: $19 \to 64 \to 32$ with ReLU and dropout 0.1.

17.4. Results

Model	Image-only Acc	+metadata Acc	Δ Acc	+meta F1
EfficientNet-B0	0.8796	0.8337	-4.6 pp	0.768
ResNet50	0.8636	0.8782	$+1.5$ pp	0.810
DenseNet121	0.8543	0.8550	$+0.1$ pp	0.793
Swin-Tiny	0.8490	**0.8982**	**+4.9 pp**	**0.871**
Ensemble	0.8949	**0.9049**	+**1.0** pp	**0.8728**

Three breakthrough findings

(1) Swin + metadata alone beats the image-only ensemble! 89.82% > 88.89%. Self-attention integrates metadata more effectively than CNNs in this setting.
(2) EB0 loses 4.6 pp. The smallest model (5.3M) does not have enough capacity for a meta encoder without sacrificing image features.
(3) The ensemble reaches 90.49% / 0.873 / 0.984 AUC — beating Pacheco & Krohling 2020 (89.0% / 0.83), which is the strongest published image+metadata fusion on ISIC 2018.

17.5. The FP16 bug story

Bug "AMP FP16 softmax NaN" — a lesson

Initially train_meta.py ran softmax under AMP autocast → FP16 probability tensors. When saved to CSV with pandas `.round(6)`, FP16 lacked the precision → **the entire column became NaN**. Per-model metrics were fine (computed in memory), but the ensemble combined NaNs → accuracy 3.26% (below random!).
Fix: cast `logits.float()` before softmax. Lesson: *always* use float32 for probabilities when writing them to disk.

18. Out-of-distribution evaluation (Week 3) — smartphone photographs

18.1. Why OOD evaluation?

Plain-English Explanation

Testing on the same distribution as training (ISIC 2018 dermoscopy) does not prove the model is deployable in the real world. I evaluated the image-only ensemble zero-shot on 2,298 smartphone images from PAD-UFES-20 (Brazil) — a completely different acquisition modality from dermoscopy.

18.2. Mapping 6 PAD-UFES → 5 ISIC

- ACK → akiec
- SCC → akiec (Pacheco clusters it with AKIEC)
- BCC → bcc
- SEK → bkl
- NEV → nv
- MEL → mel

Class	Precision	Recall	F1	AUC	Support
akiec	0.543	0.145	0.229	0.614	922
bcc	0.611	0.421	0.499	0.733	845
bkl	0.175	0.268	0.212	0.633	235
mel	0.057	0.308	0.097	0.684	52
nv	0.258	0.725	0.380	0.811	244
Accuracy 32.46% (vs. 88.89% in-distribution: -56 pp)					

18.3. Results

How to read the OOD result correctly

A 56 pp drop is NOT a disaster. It is the standard cost of zero-shot transfer from dermoscopy to smartphone (literature reports the same 50–60 pp range).

- Per-class AUC stays in 0.61–0.81 $\Rightarrow$ probability ranking transfers *partially*, the model is not random.
- NV recall 0.72 $\Rightarrow$ classical prior collapse onto the ISIC majority class.
- MEL precision 0.06 $\Rightarrow$ the model over-calls melanoma on smartphone photos $\Rightarrow$ clinically dangerous if deployed without adaptation.

Honest conclusion: the pipeline is not ready for consumer-grade smartphone photographs. Closing the gap is an **engineering exercise** (domain adaptation, in-domain fine-tuning, style transfer), *not a research mystery*.

19. Comparison with state-of-the-art

Work	Year	Method	Acc	Macro F1
Codella et al.	2018	ResNet-152	85.9	0.78
Gessert (ISIC winner)	2018	Ensemble EB_n + SENet	88.5	0.82
Mahbod et al.	2020	Multi-scale CNN ensemble	88.7	0.83
Pacheco & Krohling	2020	RN50 + metadata fusion	89.0	0.83
This thesis (image only)	2026	4-model ensemble, 5 seeds	89.93±0.66	0.852±0.010
This thesis (+meta)	2026	**4-model + metadata**	**90.49**	**0.873**

20. Honest contribution — DO NOT over-claim

Say this plainly at defense

No technique in this thesis is novel. Every component has prior literature:

- Pacheco-style metadata fusion = exactly Pacheco & Krohling 2020.
- PAD-UFES-20 OOD = Pacheco himself published the dataset.
- Soft voting, Mixup, CutMix, Label Smoothing = industry standards.
- CNN+Transformer ensembles = many published papers 2022–2024.

The real contributions:

1. **Transparent integration** — open source code, YAML configs, 5 reproducible seeds, comprehensive study guide. Rare among SOTA papers.
2. **Quantitative Grad-CAM** — focus_ratio, peak_distance measured on the test set. Most papers stop at qualitative heatmaps.
3. **Statistical robustness** — multi-seed + McNemar with Fisher's method. Most ISIC papers use a single seed and skip significance tests.
4. **Honest OOD reporting** — the 56 pp drop is reported openly. Many papers quietly hide OOD failures.
5. **Mild gain over Pacheco 2020** — 90.49% vs 89.0% thanks to a CNN+Transformer ensemble. This is an *increment*, not an *innovation*.

21. Ten hard questions and how to answer

Q1: Why soft voting and not stacking?

Three reasons: (1) Stacking needs a second val set $\rightarrow$ overfits on small data. (2) Hard voting discards probability information. (3) Soft voting already reaches 89.49% with McNemar $p < 10^{-7}$ vs. the strongest single model, leaving no clear headroom for stacking to beat.

Q2: How do you know the Grad-CAM is "correct"?

I measure four quantitative metrics: focus_ratio, peak_distance, entropy, coverage_75. All four models achieve focus_ratio > 0.6 and peak_distance < 0.2. I caught the `model.bn2` bug because of the quantitative measurements — by visual inspection alone the mistake would have slipped through.

Q3: What is the difference between Macro F1 and balanced accuracy?

Balanced accuracy = mean of per-class recall $\rightarrow$ only penalises false negatives. Macro F1 = mean of per-class F1 $\rightarrow$ penalises both false positives and false negatives. I chose F1 because, clinically, both error types have costs (missing MEL is dangerous, false alarms cause unnecessary biopsies).

Q4: Why does Swin overfit faster?

(1) Lack of inductive bias — Transformers must learn locality and translation invariance from the data. (2) 28M parameters / 7K images = 4,000× — extremely overfit-prone. (3) Quadratic self-attention can memorise specific spatial structures. I mitigated with LR 1e-4, weight decay 5e-2, 5-epoch warmup.

Q5: What about images outside the 7 classes (acne, scars)?

The model is *forced* to assign one of 7 classes; there is no "other" class. I handle this two ways: (1) a confidence threshold of 50% triggers a low-confidence warning in the app. (2) high predictive entropy is an OOD signal. A production system needs a dedicated OOD detector (Mahalanobis distance, Energy score) — I did not implement that, it is future work.

Q6: How many melanoma false negatives are there?

167 true MEL images, the ensemble correctly identifies 124 → **43 misses** (recall 0.74). 35 are predicted as NV, 8 as BKL. Threshold tuning with Youden's J lowers the MEL threshold 0.143 → 0.100, which raises recall 0.74 → 0.83 at the cost of a slight rise in false positives.

Q7: Why not use Focal Loss?

I tried it; the code lives in `losses.py`. Focal Loss combines poorly with Mixup/CutMix because mixed labels like $[0.7, 0.3]$ are not one-hot. I chose Label Smoothing, which is simpler and combines natively with Mixup, and the result 0.845 F1 is already strong enough.

Q8: What is the ECE?

EB0: 0.041 → 0.018 after temperature scaling. Swin highest at 0.067 → 0.024. Ensembling models that have already been calibrated lowers ECE further. After temperature scaling, the probabilities are trustworthy enough for clinical decisions.

Q9: White-skin bias — what about Vietnam?

I acknowledge this limitation. HAM10000 is mostly Fitzpatrick I–III (Austria/USA). The OOD result on PAD-UFES-20 (Brazil, more Fitzpatrick III–IV) shows a 56 pp drop — partly due to skin-tone shift. Remedies: (1) fine-tune on Vietnamese data, (2) Fitzpatrick-aware hue/saturation augmentation, (3) group fairness loss.

Q10: What else is needed for production?

(1) CE marking / FDA 510(k). (2) In-domain data (500–5K hospital images). (3) Open-set detector. (4) Clinical UI integrated with PACS/EHR and an audit log. (5) Post-deployment monitoring (data drift, performance). (6) Clear liability assignment when errors occur. I do not claim production-readiness — this is a research prototype.

22. Defense kit checklist

22.1. Hardware

- Primary laptop + charger (check battery is full).
- USB-C / HDMI adapter for the projector.
- External mouse or presentation clicker.
- Backup USB drive containing: slides VN + EN + thesis + study guide + 5 demo screenshots.

- One printed copy of slides 4-up.

22.2. Software and demo

- **15 minutes before defense:** `streamlit run app.py` → `http://localhost:8501`.
- Set theme (Dark / Light) via menu ⋮ → Settings → Theme.
- Upload one demo image, view Grad-CAM for all 4 models → warms up the cache.
- Prepare 3 demo images: 1 clear MEL, 1 benign NV, 1 low-confidence case to demonstrate the warning.

22.3. Backup plans

- Streamlit crashes → use the 5 demo screenshots embedded in the slide.
- Projector does not recognise the laptop → open the slides PDF from USB on the room's machine.
- No internet → app.py runs 100% offline.

22.4. One hour before

- Test on the actual projector (if possible).
- Listen to the last recorded rehearsal.
- Light snack (banana, almonds).
- Restroom.
- Phone → airplane mode.

23. One-page summary (print and carry)

Project at a glance

Problem: 7-class skin lesion classification, ISIC 2018, 58:1 imbalance.
Pipeline: fine-tune 4 ImageNet-pretrained backbones (EB0, RN50, DN121, Swin) + Weighted Sampler + Mixup(α=0.2) + CutMix(α=1.0) + Label Smoothing(ε=0.1) + AdamW + cosine warmup + FP16.
Soft-voting ensemble of 4 models $\Rightarrow$:

- 1 seed: **89.49% Acc / 0.845 F1 / 0.980 AUC**
- 5 seeds: **89.93** $\pm$ **0.66% / 0.852 / 0.982**
- McNemar Fisher $p < 10^{-7}$ vs. every single model.

+ Metadata fusion (19-dim: age + sex + localisation) $\Rightarrow$:

- Ensemble: **90.49% Acc / 0.873 F1 / 0.984 AUC** (beats Pacheco 2020).
- Swin + metadata 89.82% — alone beats the image-only ensemble.

OOD on PAD-UFES-20: 32.46% (56 pp drop), per-class AUC 0.61–0.81.
Ablation: Mixup/CutMix is the most important (-4.51% F1 when removed).
Quantitative Grad-CAM: focus_ratio > 0.6, peak_distance < 0.2 across all 4 models.
Limitations: poor smartphone OOD, no metadata $\rightarrow$ Vietnam (skin-tone bias), narrow 7-class scope.
Future work: domain adaptation, open-set detection, active learning loop.
Honest contribution: no novel techniques — the contributions are *transparent integration, quantitative Grad-CAM, statistical robustness, honest OOD, and a mild lift over Pacheco through architectural diversity*.

Read this guide three times before defense. Master the 10 Q&As. Memorise the one-page summary. You are ready.